

Shelter-in-Place Guidance

Immediate Fallout Protection — Fully Cited Reference Document

This document summarizes shelter-in-place, fallout protection, decontamination, and emergency timeline protocols sourced directly to U.S. federal agency guidance including CDC, EPA, FEMA, DHS, and Ready.gov. All 26 citations are numbered inline and listed with full URLs in the References section. Always follow official emergency management instructions during an actual event. Download the latest version at: nuclearthreatwatch.org/resources/printables/shelter-in-place-guidance.pdf

1 — SHELTER-IN-PLACE GUIDANCE

- **When to Shelter:** Enter shelter immediately upon a nuclear or radiological alert or explosion. CDC advises get inside as soon as you can and stay there until officials say it is safe. [1][2] Fallout begins arriving approximately 10 minutes after a ground detonation — seek shelter before fallout descends. [3]
- **Building Type:** Choose a sturdy, well-constructed building. EPA and CDC emphasize brick, concrete, or underground structures such as basements, subways, and underground garages as the safest shelters. [4][5] Any indoor building is better than outdoors, but multi-story concrete or brick buildings give extra protection. [6][7]
- **Basement vs Upper Floors:** The basement or underground level is best, offering maximum shielding from earth and concrete. [4][8] If no basement is available, choose a windowless interior room in the middle of the building, away from exterior walls and the roof. [8][9] Stay on lower floors if possible.
- **Shielding Principles:** Put as many layers of dense material between you and the outside as possible. CDC's ALARA principle says use time, distance, and shielding — go deep inside and add walls. [10][6] EPA notes concrete walls act as a protective barrier against radiation. [5] In practice: close shutters, seal cracks, and stack books or sandbags for extra shielding.
- **Lockdown Actions:** Once sheltered, seal the environment. CDC instructs: close and lock all windows and doors, turn off outside air vents, fans, air conditioners, and fireplace dampers. [2] Stay away from exterior walls and roof. Cover your mouth and nose if dust-laden air is unavoidable. [2][7]
- **Shelter Duration:** Stay sheltered at least 24 hours, or until authorities say it is safe. Both CDC and EPA stress that walls block radiation and radioactive materials become weaker over time — staying 24 or more hours inside can protect you until fallout decays to safer levels. [11][8]
- **When to Leave:** Do not leave shelter until told by officials. EPA and CDC advise waiting for an all-clear from public safety authorities. [8][7] Only official evacuation orders or reentry guidelines will confirm it is safe to exit.

● Safety Tip

The first 24 hours are the highest-risk window. Fallout radiation intensity drops by roughly 90% within the first 7 hours after detonation (the 7-10 rule). Staying inside during this window — even in an ordinary wood-frame home — dramatically reduces your total dose compared to being outdoors.

2 — FALLOUT PROTECTION METHODS

- **Time:** Minimize time exposed. Radiation dose accumulates with time outdoors. Follow ALARA: finish tasks quickly and stay sheltered as long as needed. [12] Fallout gamma radiation decays by approximately half each hour — this is the basis of the 7/10 rule.

- **Distance:** Maximize distance from fallout sources. EPA and CDC recommend putting as many walls between you and the outside as possible. [10][6] The farther from the blast, the lower the dose.
- **Shielding:** Use dense shielding — concrete, brick, earth, or heavy debris. [5][6] In a high-rise, go to interior floors rather than the top. Filled bookshelves, stacked water containers, and clothing piles between you and exterior walls all add meaningful protection.
- **Interior Room Guidance:** EPA and CDC advise sheltering in basements or inner rooms without windows. [8][9] A center room maximizes distance from fallout. Even 8 to 10 feet of earth provides excellent gamma shielding.
- **Basement Protection Factor:** EPA notes basements can act as fallout shelters with high protection factors — often 10x or more dose reduction compared to outdoors. [5][4] Stock the basement with supplies if you live near likely targets.
- **Building Materials:** Concrete, brick, and earth are best; wood and metal protect less. [5] Fill gaps under doors with wet towels and use layers of clothing or book stacks on walls for extra reduction.

3 — DECONTAMINATION PROCEDURES

- **Removing Clothing:** Remove all outer clothing immediately after potential exposure. Removing outer garments can eliminate up to 90% of radioactive contamination from your body. [13][14] Handle clothes carefully — do not shake them — and place in a sealed plastic bag away from people.
- **Washing Skin and Hair:** Shower with soap and water as soon as practical. [15][14] Focus on face, hands, and hair. Scrub gently. Do NOT use alcohol wipes or hand sanitizer — these are ineffective for decontamination. [15] Change into clean indoor clothing after washing.
- **Contaminated Items:** Keep contaminated clothing and wipes in sealed plastic bags. Do not bring radioactive dust into a clean space — shoes and gear stay at the door. Wipe hard surfaces with damp cloths.
- **Preventing Inhalation:** Avoid breathing fallout dust. If you must go outside briefly, cover your mouth and nose with an N95 mask or damp cloth. [16] Do not run fans that draw outside air.
- **Food and Water Safety:** Drink only sealed water and eat only covered or packaged food. [17][18] Wipe down canned or bottled food that was outdoors before opening. Do not eat garden produce or unpackaged food until authorities test it.
- **Pets and Animals:** Bring pets indoors immediately. Wash with mild shampoo to remove fallout dust. [19][20] Use clean water — not household wipes — on animals. Feed pets sealed food.

● Safety Tip

Have your decontamination supplies staged before you need them. A bin near your entry point containing garbage bags, disposable gloves, and a change of clothes per household member lets you complete the clothing-removal step without entering your shelter room while still contaminated.

4 — EMERGENCY TIMELINE AFTER FALLOUT

TIME WINDOW	GUIDANCE
0–10 Min (CDC/FEMA)	Get inside immediately. Drop behind cover if you witnessed a blast, then move to a building. Remove shoes and outer clothing before entering. Decontaminate skin if possible. [2][14]
~10 Min (FEMA/Ready.gov)	Fallout begins arriving approximately 10 minutes after the burst. If outdoors, move rapidly to a sturdy building or basement. [3][6]

TIME WINDOW	GUIDANCE
First Hour (EPA/CDC)	Peak hazard period for fallout gamma radiation. Radiation falls by roughly half each hour. EPA notes radiation becomes significantly less dangerous during the first 24 hours. [21] Remain sheltered.
First 24 Hours (CDC/EPA)	Stay sheltered continuously. CDC specifically says stay inside at least 24 hours to allow the worst of fallout to decay. [11] Only leave for an immediate life threat such as fire, or an official evacuation order.
24–72 Hours (EPA/CDC)	After one day, radiation drops substantially — roughly 90% by 7 hours after fallout. Officials will assess areas and may begin controlled evacuations. Follow public instructions. [21]
Long-Term (EPA/FEMA)	Recovery shifts to cleanup. EPA Protective Action Guide (PAG) guidelines direct decontamination plans. [25][26] Authorities measure radiation in food and water and reopen areas gradually. FEMA and CDC provide health monitoring support. [22][27]

5 — SUPPORTING REFERENCES & CITATIONS (26 SOURCES)

All references below are U.S. government publications or official agency websites. Citations are numbered as they appear inline throughout this document.

- [1] Be Ready for Radiation Emergencies | CDC
<https://www.cdc.gov/radiation-emergencies/features/be-ready.html>
- [2] What to Do: Get Inside | Radiation Emergencies | CDC
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [3] Radiation Emergencies — Michigan Ready (reflects FEMA/Ready.gov guidance)
<https://www.michigan.gov/miready/be-informed/radiation-emergencies>
- [4] What to Do: Get Inside | CDC (basement/underground shelter)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [5] RadTown Radiation Protection Activity 4: Buildings as Shielding | EPA
<https://www.epa.gov/radtown/radtown-radiation-protection-activity-4-buildings-shielding>
- [6] What to Do: Get Inside | CDC (multi-story building protection)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [7] Preparing for a Radiation Emergency | CDC
<https://www.cdc.gov/radiation-emergencies/safety/index.html>
- [8] Radiation Emergencies and Preparedness | EPA
<https://www.epa.gov/radtown/radiation-emergencies-and-preparedness>
- [9] What to Do: Get Inside | CDC (interior room, no windows)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [10] Radiation Safety | Radiation and Your Health | CDC
<https://www.cdc.gov/radiation-health/safety/index.html>
- [11] What to Do: Get Inside | CDC (24-hour minimum shelter duration)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [12] Radiation Safety — ALARA Principle | CDC
<https://www.cdc.gov/radiation-health/safety/index.html>
- [13] Radiation Emergencies — Michigan Ready (clothing removal / 90% reduction)
<https://www.michigan.gov/miready/be-informed/radiation-emergencies>
- [14] What to Do: Get Inside | CDC (decontamination after exposure)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [15] Radiation Emergencies — Michigan Ready (no alcohol wipes guidance)
<https://www.michigan.gov/miready/be-informed/radiation-emergencies>

- [16] What to Do: Get Inside | CDC (covering mouth and nose outdoors)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [17] Radiation Emergencies and Preparedness | EPA (food and water safety)
<https://www.epa.gov/radtown/radiation-emergencies-and-preparedness>
- [18] Radiation Emergencies — Michigan Ready (food and water)
<https://www.michigan.gov/miready/be-informed/radiation-emergencies>
- [19] What to Do: Get Inside | CDC (pets and animals)
<https://www.cdc.gov/radiation-emergencies/response/get-inside.html>
- [20] Radiation Emergencies — Michigan Ready (pets/animals decontamination)
<https://www.michigan.gov/miready/be-informed/radiation-emergencies>
- [21] Radiation Emergencies — Michigan Ready (radiation decay rates, first 24 hrs)
<https://www.michigan.gov/miready/be-informed/radiation-emergencies>
- [22] Population Monitoring | Radiation Emergencies | CDC
<https://www.cdc.gov/radiation-emergencies/php/population-monitoring/index.html>
- [23] A Guide to Operating Public Shelters in a Radiation Emergency | CDC/NACCHO
<https://www.cdc.gov/radiation-emergencies/media/pdfs/2024/04/operating-public-shelters-1.pdf>
- [24] Radiation Emergencies and Preparedness | EPA (RadTown public guidance)
<https://www.epa.gov/radtown/radiation-emergencies-and-preparedness>
- [25] Protective Action Guides (PAGs) | EPA
<https://www.epa.gov/radiation/protective-action-guides-pags>
- [26] Protective Action Guides Manual | EPA (cleanup and drinking water safety)
<https://www.epa.gov/radiation/protective-action-guides-pags>
- [27] Community Reception Center (CRC) Toolkit | CDC
<https://www.cdc.gov/radiation-emergencies/php/population-monitoring/index.html>

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